

PROFESSIONAL PROFILE

Aerospace and Mechanical Design Engineer with 20+ years of experience across aircraft, spacecraft, launcher and advanced vehicle programmes. Strong in CATIA V5 design, CAD/PDM discipline, BOM/release documentation, industrialisation, production support, non-conformity resolution and practical design decisions under schedule pressure. Reliable interface between engineering, manufacturing, quality, suppliers and programme stakeholders.

TARGET ROLES

Mechanical Design Engineer	CATIA V5 Design Engineer	Aerospace Mechanical Design Engineer
Design & Development Engineer	Liaison / Production Support Engineer	Mechanical Integration Engineer
Industrialisation Engineer	Configuration / Change Engineer	Project Engineer - Mechanical Hardware

CORE COMPETENCIES

CAD / CAx	CATIA V5/V4 including GSD, DMU and installation workbenches; SolidWorks; AutoCAD; 3D parts, assemblies, installation models, 2D drawings and GD&T.
PLM / PDM / ERP	ENOVIA 3DEXPERIENCE, SmarTeam, VPM, Windchill, Teamcenter exposure, SAP, Primes 3, EPD Optegra, Sprint; BOM/KSL, kits and release packages.
Industrialisation	Prototype-to-production development, manufacturing readiness, DFM/DFA, production release, build-issue resolution and design changes for manufacturability.
Production support	Shopfloor liaison, non-conformity assessment, NRB/MRB-style workflows, RCA discussions, DQN/Special Instructions, deviations and corrective actions.
Hardware focus	Aerospace and launcher structures, composite/CFRP, machined parts, sheet metal, brackets, mechanisms, interiors and electromechanical installations.
Interfaces	Mechanical-electrical integration, sensor installation, pressure/temperature transducers, calorimeters, harness routing, packaging and space-claim coordination.

PROFESSIONAL EXPERIENCE

Professional Development and Career Transition | Zurich, Switzerland | 10/2024 - Present

- Following the planned completion of a fixed-term launcher programme assignment, focused full-time on securing the next contract or permanent engineering role in Switzerland / nearby Europe.
- Continued professional development in German for technical workplaces, systematic problem solving, conflict management, AI-supported technical research/writing and project management; IPMA Level D preparation ongoing.
- Maintained a targeted engineering profile around CATIA design, industrialisation, production support, quality interface and project engineering for aerospace, defence and advanced manufacturing roles.

Mechanical Design Engineer / Mechanical Design Lead - Launcher Structures | Beyond Gravity / RUAG Space via Lewry TSE | Zurich, Switzerland | 11/2018 - 09/2024

- Developed and industrialised the ULA Vulcan Centaur Heat Shield structure from concept and demonstrators through manufacturing and assembly readiness.
- Created and maintained CATIA V5 parts, assemblies, installation models, 2D drawings, BOM/KSL data, kits and SAP-based technical documentation.
- Supported production through non-conformity assessment, NRB/RCA actions, design changes and practical build-issue resolution to improve manufacturability, quality, cost and delivery flow.
- Designed mechanical-electrical integration for pressure/temperature transducers, sensors, calorimeters and harness routing through structural areas.
- Coordinated design reviews and daily technical topics with thermal/stress, systems, M&P, manufacturing, quality, safety, tooling suppliers and international teams in the USA and Portugal.

PROFESSIONAL EXPERIENCE - CONTINUED

Senior Design Engineer - Aircraft Interior / Retrofit | **Aerolite AG via Lewry TSE** | Ennetburgen, Switzerland | 04/2018 - 11/2018

- Supported medical conversion of the Pilatus PC-24 RFDS under CS-23 constraints, including red-mark review of drawings/BOMs and production release support.
- Re-designed machined parts, sheet-metal brackets, hinges and interior components in SolidWorks for manufacturability under tight deadlines.
- Integrated electronic equipment into liners and interior structures and coordinated updated 3D data with Pilatus.

Primary Structure Design Engineer - Chassis / BIW / Mechanisms | **AeroMobil s.r.o.** | Bratislava, Slovakia | 07/2016 - 04/2018

- Designed chassis, body-in-white structures, body systems, composite/CFRP structures and mechanisms from prototype to production readiness.
- Performed packaging and concept studies balancing aerodynamics, styling, space claim, manufacturability, assembly feasibility and system requirements.
- Developed folding-wing attachment concepts, engine mounting interfaces and control surface mechanisms in CATIA V5 GSD.

Senior Mechanical Design Engineer - Helicopter and Spacecraft Integration | **Specialist Aviation Services / Airbus Defence & Space** | UK | 04/2015 - 06/2016

- Designed AW169 helicopter interior modifications for medical equipment and TETRA radio systems, including brackets, supports, installation drawings and BOMs.
- Defined SES-14 satellite equipment layouts, baseplates, brackets, waveguide/coax routing, 3D installation drawings, GD&T and ECN implementation in CATIA V5/VPM.
- Worked on clean-room installation feasibility, configuration control and controlled technical documentation for space hardware.

Liaison Engineer - Production Support / Design Changes | **Jet Aviation / General Dynamics** | Basel, Switzerland | 06/2011 - 06/2012

- Supported Airbus A319 CJ VVIP completion as shopfloor liaison engineer, resolving manufacturing and installation issues across structures, cabin, HVAC, water and electrical interfaces.
- Developed practical design and installation solutions during production and final assembly, including non-conformity disposition and controlled design changes.
- Worked in EASA Part 145 / Part 21G / Part 21J contexts in support of STC and delivery requirements.

Mechanical Design Engineer - Final Assembly and Earlier Aerospace Roles | **Airbus Military / Tods Aerospace / Airbus UK / EADS-CASA** | Spain and UK | 09/2005 - 06/2011

- Supported early A400M Final Assembly Line units MSN001-MSN007, resolving non-conformities from Task 100 design assessment through implementation.
- Produced DQNs, Special Instructions, technical logbook statements and limitation sheet coordination to keep test and production activities moving.
- Earlier CATIA V4/V5 and PDM work on Lynx helicopter panels, A380 HVAC, A400M flight-test installations, P3 Orion stowage, A330 MRTT ARO console and A380 harness/BOM/ECN topics.

EDUCATION, LANGUAGES AND SELECTED TRAINING

Education	Technical Aeronautical Engineer / Aerospace Engineering, Universidad Politecnica de Madrid, Spain.
Languages	Spanish native English C2 German B2.
Work status	EU citizen Swiss B permit Available immediately / by agreement.
Selected training	EASA / Part 21 Subpart J seminar; conflict management certificate; IPMA Level D preparation ongoing.